

Energy and efficiency: On India and greater energy efficiency mandates

India must reduce its power consumption by increasing efficiency

Despite **robust** growth in electricity generation over the past two **decades**, with **rapid** additions of **renewable energy** in the past five years, India has been unable to **meet** its peak power demand, with the **deficit widening** from 0.69% in FY20 to about 5% in FY24. This reveals **constraints** in the supply of power — new power production is time consuming, especially if fossil-fuel based, even as India attempts to **integrate** renewable power into the **power grid**. Therefore, India must focus on enhancing energy efficiency **holistically** to reduce power demand, also the quickest and least expensive way to **address** rising power demand and climate change. This year **marks** a decade of India's **groundbreaking** energy efficiency scheme, UJALA, which has helped **bring down** the price of energy efficient light emitting diode (LED) bulbs from about ₹500 a decade ago to ₹70, enabling its widespread home use. The scheme succeeded as another public energy efficiency measure was **baked** into the initiative — the Street Lighting National Programme, which **led to** the installation of over 1.34 crore LED lamps across urban local bodies and gram panchayats, and reducing peak demand by over 1,500 MW. As of January 2025, the government has distributed about 37 crore LED bulbs and enabled the sale of about 407 crore more.

LED bulbs consume half the amount of power of compact fluorescent lamps, while **incandescent** light bulbs require nine times the power that LEDs consume, translating into **considerable** cost savings for Indian homes. But estimates also suggest that the UJALA scheme alone has helped India save over \$10 billion and avoided building over 9,500 MW of new generation capacity, which is the equivalent of 19 new coal-fired 500 MW power plants. **Indeed**, there are other energy efficiency measures that India has taken **following** the **enactment** of the Energy Conservation Act, 2001. The International Energy Agency **states** that between 2000 and 2018, energy efficiency improvements enabled India to avoid an additional 15% of energy demand and 300Mt of CO₂ emissions. But with India's rapid **urbanisation** in the past two decades and rising **per capita** energy consumption to meet cooling **needs** as summers get hotter, peak power demand reached 250 GW last year. India is today the third largest power consumer globally, after China and the United States. **Moreover**, **70%** of its energy output **continues** to be from coal and India has plans to add another 90 GW of coal-based capacity by 2032. What is needed now is greater energy efficiency mandates across sectors such as buildings, home **appliances** and the country's **sprawling** MSMEs. [Practice Exercise]

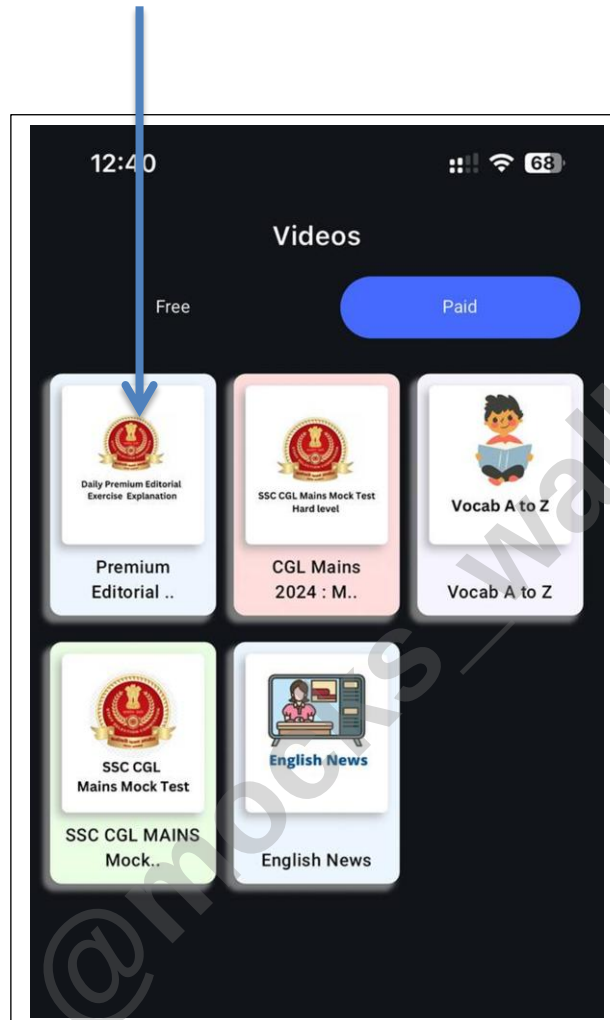
- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

1. **Mandate** (noun) – directive, order, command, instruction, decree आदेश / अनिवार्यता
2. **Robust** (adjective) – strong, resilient, sturdy, powerful, vigorous मजबूत
3. **Decade** (noun) – Period of ten year दशक
4. **Rapid** (adjective) – fast, quick, swift, speedy, accelerated तीव्र
5. **Renewable energy** (noun) – energy derived from natural sources that are replenished at a higher rate than they are consumed नवीकरणीय ऊर्जा
6. **Meet** (verb) – fulfill, satisfy, comply with, achieve, address पूरा करना
7. **Deficit** (noun) – shortage, shortfall, gap, lack, insufficiency घाटा
8. **Widen** (verb) – expand, enlarge, increase, extend, broaden बढ़ना
9. **Constraint** (noun) – limitation, restriction, obstacle, hindrance, bottleneck प्रतिबंध / बाधा
10. **Integrate** (verb) – combine, merge, assimilate, incorporate, unify एकीकृत करना
11. **Power grid** (noun) – an interconnected network that delivers electricity from power plants to consumers, including homes, businesses, and industries.
12. **Holistically** (adverb) – comprehensively, entirely, overall, thoroughly, integrally समग्र रूप से
13. **Address** (verb) – tackle, deal with, resolve, respond to, confront समाधान करना
14. **Mark** (verb) – signify, commemorate, denote, celebrate, represent चिह्नित करना
15. **Ground-breaking** (adjective) – innovative, pioneering, revolutionary, trailblazing, novel क्रांतिकारी
16. **Bring down** (phrasal verb) – reduce, lower, decrease, cut, diminish कम करना
17. **Bake** (verb) – incorporate, include, embed, integrate, build in शामिल करना
18. **Lead (to)** (verb) – result in, cause, bring about, produce, give rise to कारण बनना
19. **Incandescent** (adjective) – glowing, luminous, radiant, bright, light-producing दीप्तिमान / तेज
20. **Considerable** (adjective) – significant, substantial, marked, notable, large पर्याप्त / महत्वपूर्ण
21. **Indeed** (adverb) – certainly, truly, in fact, undeniably, actually वास्तव में
22. **Following** (preposition) – after, subsequent to, in the wake of, ensuing, resulting from के बाद

23. **Enactment** (noun) – passing, implementation, legislation, ratification, institution अधिनियमन / प्रवर्तन
24. **State** (verb) – declare, express, announce, report, mention कहना / बताना
25. **Urbanisation** (noun) – city development, urban growth, metropolitan expansion, town planning, urban sprawl शहरीकरण
26. **Per capita** (phrase) – per person, per individual, for each person, average per head प्रति व्यक्ति
27. **Moreover** (adverb) – besides, additionally, furthermore, in addition, also इसके अलावा
28. **Appliance** (noun) – device, equipment, gadget, machine, tool उपकरण
29. **Sprawling** (adjective) – spread-out, extensive, widespread, scattered, expansive फैला हुआ

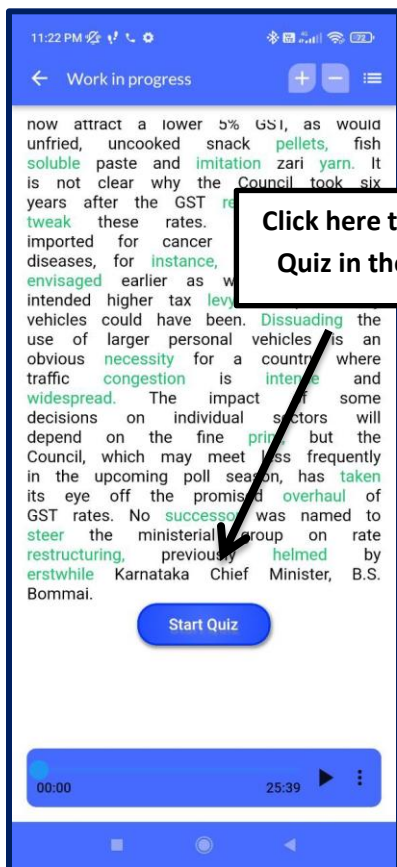
Video solution of Practice Exercise from this Editorial is exclusively available on App.



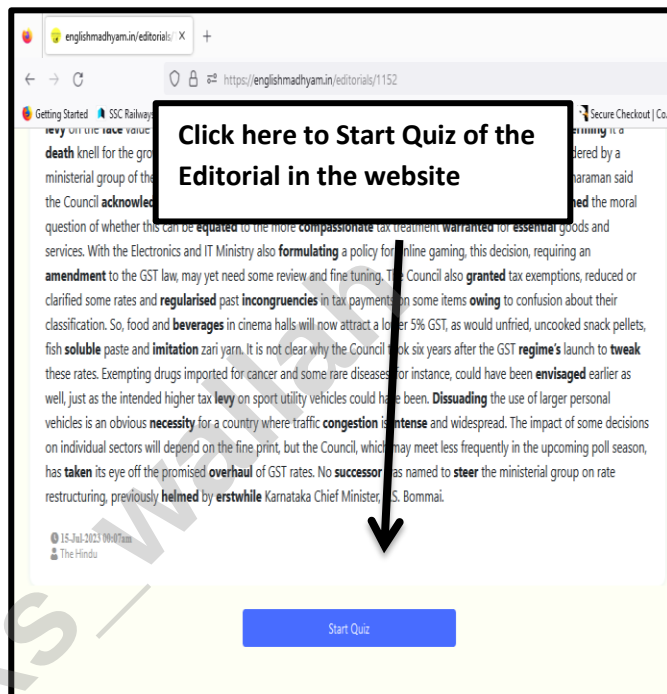
Summary of the Editorial

1. **Rising Power Demand:** India's electricity generation has grown, yet peak power shortages have increased from 0.69% (FY20) to 5% (FY24).
2. **Supply Constraints:** Meeting demand through new power generation, especially fossil-based, is time-consuming and difficult.
3. **Efficiency as a Solution:** Enhancing energy efficiency is the fastest and most cost-effective way to reduce power demand and tackle climate change.
4. **Success of UJALA Scheme:** Launched a decade ago, UJALA drastically reduced LED bulb prices from ₹500 to ₹70, enabling mass adoption.
5. **Street Lighting Programme:** Over 1.34 crore LED streetlights have been installed under this scheme, significantly lowering peak demand.
6. **LED Distribution Impact:** By January 2025, 37 crore LED bulbs were distributed and 407 crore sold, contributing to massive energy savings.
7. **Energy Savings by LEDs:** LED bulbs consume 50% less power than CFLs and 1/9th compared to incandescent bulbs.
8. **Avoided Infrastructure Costs:** UJALA alone helped avoid building 9,500 MW of power capacity — equal to 19 coal plants of 500 MW each.
9. **Financial Benefits:** The scheme has saved India over \$10 billion through reduced electricity costs and infrastructure savings.
10. **Broader Efficiency Gains:** Since the Energy Conservation Act (2001), India avoided 15% more energy demand and 300 million tonnes of CO₂ emissions (2000–2018).
11. **Growing Urbanisation:** Rapid urbanisation and hotter summers have driven higher per capita electricity use, especially for cooling.
12. **Record Power Demand:** India's peak power demand reached 250 GW last year, making it the world's third-largest power consumer.
13. **Coal Dominance:** About 70% of India's energy comes from coal, and another 90 GW of coal-based capacity is planned by 2032.
14. **Need for Broader Mandates:** Stronger energy efficiency mandates are required across buildings, appliances, and MSMEs.
15. **Path to Sustainability:** Improving efficiency is not just a climate imperative but also an economic and infrastructure necessity for India's energy future.

Practice Exercise: SSC Pattern Based



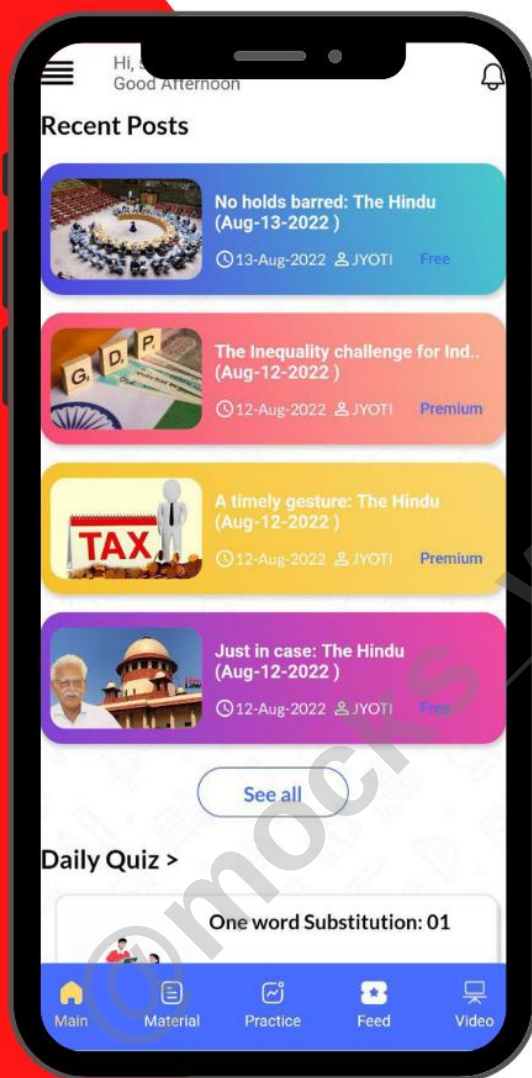
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